1 Install Wireshark on your laptop, start a capture on your main network interface, and visit any website (like instagram.com). Stop the capture and note down the protocol names you see in the packet list.

I installed Wireshark on my laptop and started capturing packets on my active network interface. I opened a website such as Instagram and then stopped the capture. The Protocol column showed different protocols used for network communication.

Common protocol names that may appear are:

- DNS — Domain Name System
- TCP — Transmission Control Protocol
- TLS — Transport Layer Security
- QUIC — Quick UDP Internet Connections
- UDP — User Datagram Protocol
- DHCP — Dynamic Host Configuration Protocol
- ARP — Address Resolution Protocol
- HTTP — Hypertext Transfer Protocol

2 Pick any 3 packets from your Wireshark capture and identify which OSI layer each packet mainly represents (for example, HTTP for Application layer, TCP for Transport layer, etc.).

Hint: Use the Protocol column and Info details in Wireshark to help you map protocols to OSI layers.

Protocol	OSI Layer	Purpose
DNS	Layer 7 – Application	Converts domain names into IP addresses.
TCP	Layer 4 – Transport	Provides reliable data delivery.
ARP	Layer 2 – Data Link (commonly mapped)	Helps find the MAC address associated with an IPv4 address on a local network.

Conclusion: Wireshark helps us inspect network packets and understand how different protocols communicate across network layers.

3 Create a colorful diagram of the OSI Model using any drawing tool (MS Paint, Canva, or even paper and upload a photo). For each layer, add a real-world example from apps you use daily (e.g., WhatsApp uses the Application layer, Instagram images use the Presentation layer, etc.).

Use the generated diagram above as a visual reference. Draw these seven layers from top to bottom in your notebook.

Layer	Name	Real-world example
7	Application	WhatsApp, Instagram
6	Presentation	Image formats, data encryption
5	Session	Maintaining communication sessions
4	Transport	TCP, UDP
3	Network	IP address, router
2	Data Link	Wi-Fi, Ethernet, MAC address
1	Physical	Network cables, radio signals

Easy trick to remember the layers (top to bottom):

Application → Presentation → Session → Transport → Network → Data Link → Physical.

4 Explain how sending a message on WhatsApp travels through at least 4 layers of the OSI Model, describing what happens at each layer in your own words.

Hint: Think about how your message is typed, packaged, sent over the internet, and received.

When I send a message on WhatsApp, it passes through different network layers.

1. Application Layer: I type a message and press the Send button in WhatsApp.

2. Presentation Layer: The data is represented in a suitable format and protected using encryption.

3. Transport Layer: TCP or another transport protocol carries the data between endpoints.

4. Network Layer: IP addresses help route the data toward the destination.

5. Data Link Layer: The data is placed into frames and sent over Wi-Fi or Ethernet.

6. Physical Layer: The data travels as radio signals or electrical signals through the network.

Finally, the message reaches WhatsApp's infrastructure and is delivered to the receiver's device. The receiving device processes the data in reverse order so the message can appear on the screen.

Conclusion: The OSI model helps us understand how data travels from one device to another through different network layers.